

1. Administrative & Study Identification

Full Study Title: Study to Assess the Efficacy, Pharmacokinetics, Safety and Tolerability of Iptacopan in Pediatric Patients With Primary IgAN **Short Title/Acronym:** Iptacopan in Pediatric Primary IgAN **Protocol Number:** Not Available **NCT Number:** NCT06994845 **Posting ID:** 707851423148424182 **Sponsor Name:** Novartis (Novartis Pharmaceuticals) **Investigational Product:** Iptacopan hydrochloride **Phase of Development:** Phase III **Trial Status:** Not Yet Recruiting **Medical Monitor:** Novartis Clinical Operations / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To determine proteinuria reduction, pharmacokinetics (PK), safety, and tolerability (including cardiovascular surveillance) of iptacopan in primary immunoglobulin A nephropathy (IgAN) pediatric patients aged 2 to <18 years. **Secondary Objectives:** To evaluate pharmacokinetic parameters in plasma, alongside clinical measures such as eGFR slopes and time to composite kidney failure events, and longitudinal changes in proteinuria (UPCR).

2.2 Background & Rationale A growing body of evidence links the overactivation of the complement alternative pathway with glomerular inflammation and fibrosis in complement-mediated kidney diseases (CMKDs) like IgAN. Iptacopan is an oral, highly selective Factor B inhibitor that exerts a proximal effect in the complement cascade, thereby targeting the underlying pathogenesis to prevent renal cell damage. This trial seeks to evaluate its efficacy and safety specifically within the pediatric population to expand treatment options beyond standard supportive care.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N:** 24 pediatric patients **Number of Sites:** Multicenter **Estimated Study Start:** March 2026 **Estimated Primary Completion:** Late 2027 (52-week treatment period)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female participants 2 to < 18 years of age as of Day 1. 2. eGFR ≥ 30 mL/min/1.73m² calculated using the modified Schwartz formula at Screening and confirmed during the Run-in Period. 3. Kidney biopsy-proven primary IgAN, with biopsy performed within 3 years of Screening showing < 50% tubulointerstitial fibrosis and < 25% crescents. 4. Minimum body weight for participants in Cohort 1 is 35 kg at Screening and confirmed at Baseline. 5. Proteinuria due to primary IgAN assessed by UPCR ≥ 1 g/g despite treatment with a maximum tolerated dose of ACE inhibitor/ARB for at least 120 days prior to Day 1.

4.2 Exclusion Criteria 1. Evidence of significant urinary obstruction, difficulty voiding, or chronic kidney disease other than IgAN. 2. Current acute kidney injury (AKI) defined by AKIN criteria within 4 weeks of screening. 3. Presence of rapidly progressive glomerulonephritis (RPGN) defined by a 50% decline in eGFR within 3 months. 4. Previous treatment with immunosuppressive or immunomodulatory agents within 120 days (180 days for rituximab) prior to the first dose. 5. Clinically significant echocardiogram/ECG abnormalities (e.g., LVEF < 50%) indicating cardiovascular risk.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Oral administration of iptacopan hydrochloride. Cohort 1 (12 to < 18 years; weight ≥ 35 kg) will consist of 16 participants. Cohort 2 (2 to < 12 years) will consist of 2 groups of 4 participants (8 total).

Route of Administration: Oral **Duration of Treatment:** 52 weeks

5.2 Concomitant & Prohibited Therapy Permitted: Stable supportive care including max tolerated ACE inhibitors or ARBs. **Prohibited:** Concurrent immunosuppressive agents, other immunomodulators, or anti-complement therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -77 to -14 (Up to 11 Weeks)	Informed Consent, Medical History, eGFR/UPCR confirmation, Vaccination verification (meningococcal/pneumococcal)
Run-in Period	Day -14 to 0 (2 Weeks)	Baseline assessments, including two UPCR FMV measurements required prior to Day 1
Baseline	Day 1	Enrollment, First Dose, Baseline labs
Interim	Up to Week 38	Safety/CV surveillance, PK sampling, UPCR (FMV) collection
End of Study (EOS)	Week 52	Final Efficacy Assessment, Exit Interview, IP Accountability
Follow-up / REP		

| Post-Week 52 (1 Week) | 1-week safety follow-up or entry into open-label Rollover Extension Program (REP) |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Log-transformed ratio to Baseline in Urine Protein-to-Creatinine Ratio (UPCR) based on first morning void (FMV) at Week 38. Safety, tolerability, and PK profiles are also co-primary measures. **Measurement Method:** Central Laboratory Analysis of FMV samples and structured clinical safety monitoring.

7.2 Secondary & Exploratory Endpoints 1. Pharmacokinetic Parameter Cmax in Plasma. 2. Pharmacokinetic Parameter AUClast in Plasma. 3. Pharmacokinetic Parameter AUCtau in Plasma. 4. Annualized total eGFR slope over the treatment period. 5. Time to first composite kidney failure event.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine lab tests, cardiovascular surveillance (ECHO/ECGs), and infection monitoring. **Serious Adverse Events (SAEs):** Reported within 24 hours. Given the mechanism of action, specific monitoring for encapsulated bacterial infections (e.g., meningococcus, pneumococcus) is mandatory. **Data Monitoring Committee (DMC):** A safety review will be conducted following Cohort 1 before dosing Cohort 2.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive at least one dose) and Efficacy Set. **Sample Size Justification:** Target $N=24$ (16 in Cohort 1; 8 in Cohort 2) is deemed sufficient to model PK/PD parameters and assess preliminary safety/proteinuria reduction in the pediatric subgroup. **Handling of Missing Data:** Standard mixed-effects models for repeated measures (MMRM) or appropriate imputation methods for longitudinal continuous variables.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Centralized and onsite monitoring visits to verify source data, especially for strict inclusion (biopsy thresholds) and critical safety assessments. **Audit Trail:** Detailed electronic Case Report Form (eCRF) locks and tracking.

11. Study Identifiers & Governance

Primary ID: NCT06994845 **Posting ID:** 707851423148424182 **Registry Number:** NCT06994845 **Sponsor/Lead Organization:** Novartis **Study Title:** Study to Assess the Efficacy, Pharmacokinetics, Safety and Tolerability of Iptacopan in Pediatric Patients With Primary IgAN **Official Regulatory Title:** A Single-arm, Multicenter, Phase III Study to Assess Efficacy, Pharmacokinetics, Safety and Tolerability of Iptacopan in Pediatric Patients of 2 to <18 Years of Age With Primary Immunoglobulin A Nephropathy (IgAN)

12. Clinical Framework (IgAN Specific)

Primary Condition: Primary Immunoglobulin A Nephropathy (IgAN) **Diagnosis Threshold:** Kidney biopsy-proven primary IgAN with < 50% tubulointerstitial fibrosis and UPCR $\geq$ 1 g/g **Medical Methodology:** Alternative Complement Pathway / Factor B Inhibition **Optimization Target:** Proteinuria reduction and eGFR stabilization

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for IgAN including stable ACEi/ARB optimization **Education Protocol (HCP):** Strict adherence to prophylactic vaccination protocols against *Neisseria meningitidis* and *Streptococcus pneumoniae* **Education Protocol (Patient):** Education on signs of bacterial infections and importance of medication compliance **Quality Audit Mechanism:** Continuous PK/PD parameter tracking and adverse event auditing

14. Observation & Follow-Up Schedule

Total Duration: Up to 66 weeks (11 weeks screening + 2 weeks run-in + 52 weeks treatment + 1 week follow-up) **Onsite Visit Cadence:** Regular interim intervals through Week 38 (primary efficacy read) to Week 52 (EOS) **Remote Monitoring Frequency:** Between formal onsite visits as defined by protocol **Checklist Review Interval:** Routine medication dispensing and cardiovascular/renal check-ins

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				